3M's SCOTCH "250" HIGH ENERGY MASTERING TAPE

"Scotch" brand 250 Studio Mastering Tape, introduced first at the Los Angeles Audio Engineering Society Convention in May of 1974, is *at least* the fourth generation mastering tape in a series of studio mastering tapes developed by 3M Company for the professional recording industry.

Like those before it — No. 111, No. 202, and No. 206 — No. 250 embodies another major state-of-the-art advance in studio mastering tapes. From the 66 dB dynamic range of No. 111 to the 78 dB dynamic range of No. 250 are represented many engineering and manufacturing breakthroughs which have enabled 3M to provide the industry standard mastering tape over the years. Each succeeding development has supplanted its 3M predecessor as the industry standard.

For years, No. 111 was referenced on almost every technical data sheet published in the industry. Then in the early 60's, 3M introduced the industry's first low noise mastering tape with what had at that time an incredible 71 dB dynamic range.

Anticipating increasing industry need for greater signal-to-noise and greater dynamic range, 3M in 1970 introduced No. 206. While this tape had the same weighted NAB noise level as No. 202, the undistorted (3% third harmonic) output level of 206 is +3dB over 202, resulting in a total dynamic range of 74dB.

The latest development is No. 250. A new oxide and new coating techniques have again enabled 3M to advance the state-of-the-art in studio mastering tape. In No. 250, NAB noise drops to -72 dB and undistorted output increases another 3 dB. The net result is a mastering tape with a total dynamic range of 78 dB.

To achieve this increased output requires a greater quantity of magnetic material, hence a heavier coating weight. Achieving this, while at the same time maintaining or improving other electrical and physical properties, was the challenge in manufacturing No. 250 tape.

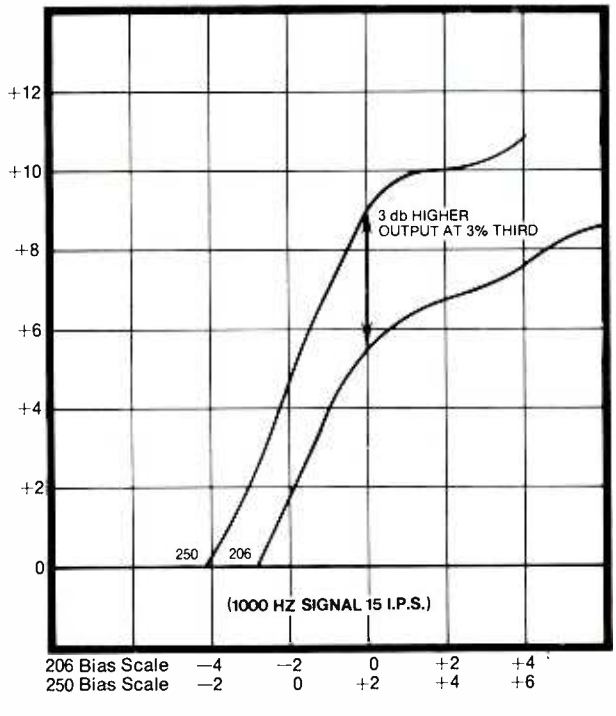
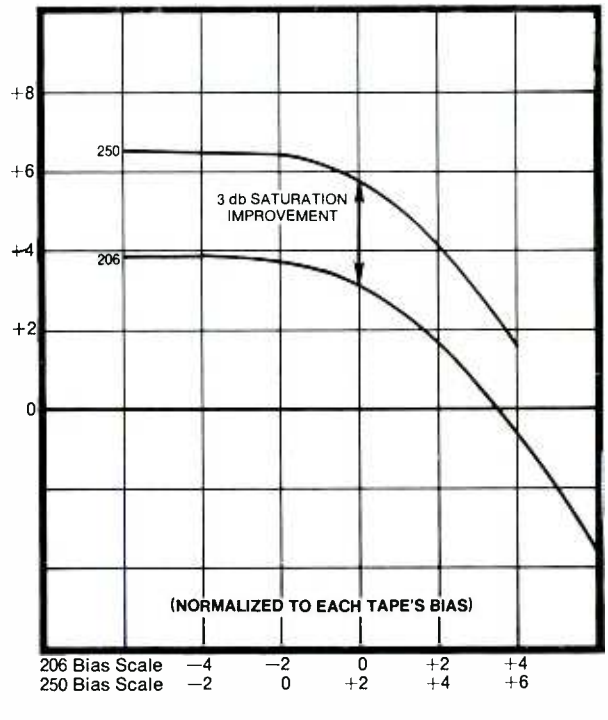
Simply increasing thickness is not the answer; 3M improved the oxide by making particles more uniform in size and needle shape, and was able to increase the density packing. At the same time, the binder was made stronger and more durable, both to accept the higher-density oxide, and to improve wear and scratch resistance. Both No. 206 and No. 250 feature a controlled wind back treatment to improve uniform high speed wind characteristics and reduce cinching and static attraction of dropout-causing contamination. Studio response to the No. 250 tape, according to 3M, has been excellent.

OPERATIONAL CHARACTERISTICS

"Scotch" 250 Studio Mastering Tape has the highest S/N ratio of any tape in the "Scotch" line - and quite probably of any line. This extension of the total dynamic range, achieved by lowering the NAB weighted noise level and increasing the level where 3%-third-harmonic distortion, is reached, allows for several operating options, depending upon the recordists' desire for additional headroom or greater signal-to-noise. These options are explained in the accompanying illustration.

Operating with a machine properly set up for "Scotch" 206 Studio Mastering Tape the operator need only adjust bias; equalization settings can remain the same. To achieve the comparative output and saturation curves accompanying this article, bias scale settings are normalized; the 250 setting is 2 dB (20 per cent) higher than that for 206.

OUTPUT AT 3% THIRD ORDER HARMONIC DISTORTION



OUTPUT AT SHORTWAVE LENGTH SATURATION AS A FUNCTION OF BIAS